Arwa Ahmed Khalil *Full-Stack Developer using .NET*

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Profile

Full-Stack .NET Developer, backend-focused, with hands-on experience across four projects spanning payments, multi-role platforms, and AI-integrated systems (Semantic Kernel, RAG). Strong in Clean Architecture, CQRS/MediatR, EF Core, and automated testing (xUnit, Testcontainers), with working Angular on the frontend. ITI-trained; currently ITI Lab Assistant.

Technical Training

03/2025 – 12/2025
Aswan

.NET Full Stack Program
Information Technology institution ITI
Completed intensive training in ASP.NET Core, Web API, Angular, EF Core, SQL Server, Git, and software engineering fundamentals through full-stack projects.

Experience

04/2026 – Present
Aswan, Egypt

ITI Lab Assistant
Information Technology Institute
- Supported students in .NET/SQL labs, debugging C#, LINQ, EF Core, and Web API issues.
- Reviewed assignments and gave technical feedback.

06/2025 – Present
Aswan, Egypt

Freelance Backend Developer
- Delivered 3 client web apps (ASP.NET Core, Angular, SQL Server), all 5-star rated.
- Gathered requirements, scoped features, and managed revisions independently.

Projects

07/2026 – 09/2026

Almalika Bakkerij: Wholesale Ordering & Delivery Platform (Client Project)
Multi-role ordering, delivery, and payments platform for a Netherlands-based bakery.
- Led backend architecture as the sole backend developer: **Clean Architecture** modular monolith (CQRS/MediatR, EF Core, SQL Server), delivered phase by phase against a written API contract for the React and React Native teams.
- Integrated **Mollie iDEAL**, cash-on-delivery, and pay-later; payment status always verified server-side via Mollie's API.
- Built an **OTP-verified delivery handshake** with offline sync, a driver cash-wallet ledger with two reconciliation flows, and audit-trailed order adjustments.
- Added FCM push notifications, delivery lanes, and Excel exports. Wrote **90+ xUnit/Testcontainers tests**, reducing execution time from **286s → 142s** via a shared fixture. Dockerized and deployed to **Azure**.

Stack: ASP.NET Core · C# · EF Core · SQL Server · MediatR/CQRS · Docker · Azure · Mollie API · Firebase Cloud Messaging · xUnit · Testcontainers

08/2026 – 09/2026

Media Portfolio Platform: Backend API (Client Project)
Bilingual (AR/EN) video-portfolio site with an admin dashboard. Backend built and tested end to end; client cancelled before launch.
- Built a **Clean Architecture** ASP.NET Core API (CQRS/MediatR, FluentValidation, EF Core on PostgreSQL) for categories, draft/publish videos, public search, and a contact form.
- Defined and **versioned the API contract** as the source of truth for a separate frontend developer.

- Secured the API with **JWT access/refresh tokens**, per-IP rate limiting, and server-side **Cloudflare Turnstile** verification; added a Hangfire job for message cleanup and race-safe analytics counters.

Stack: ASP.NET Core (.NET 10) · C# · EF Core · PostgreSQL · MediatR/CQRS · Hangfire · JWT · Docker · xUnit · Testcontainers

11/2025 – 01/2026 **Shef`ay (SmartPharmacy) -Pharmacy Management System**

AI-integrated pharmacy platform with inventory, orders, and real-time updates.

- Built secured **REST APIs** with JWT and role-based access for admins, owners, and suppliers, with **SignalR real-time updates**.
- Integrated an **AI agent (Semantic Kernel)** that reads medicine images, answers inventory queries, and places orders via tool calling, backed by a **Qdrant RAG pipeline**.
- Handled **concurrent stock deduction** safely across simultaneous orders.
- Also built a standalone **RAG application** (Angular + MongoDB vector search) for document Q&A independently during training.

Stack: ASP.NET Core · C# · Angular · EF Core · SQL Server · SignalR · Semantic Kernel · Qdrant · MongoDB · Docker · JWT

11/2025 – 11/2025 **AI-Powered RAG System (ASP.NET Core & Angular)**

Independently built during training: a RAG pipeline for AI-powered Q&A over custom documents.

- Designed the full **RAG ingestion pipeline —** parsing, chunking, embedding generation, and vector storage in MongoDB.
- Implemented **vector similarity search and retrieval** for context-aware responses, with clear separation between ingestion, retrieval, and generation layers.

Stack: ASP.NET Core Web API · Angular · MongoDB (vector) · Embedding APIs

Skills

Backend
- C#
- ASP.NET Core, Web API
- Entity Framework / EF Core
- LINQ
- Clean Architecture
- MediatR/CQRS
- JWT & Role-Based Auth

Tools & Cloud
- Docker
- Azure (Blob Storage)
- Hangfire
- Git
- SignalR
- Firebase Cloud Messaging
- Mollie API

Frontend & AI
- Angular
- TypeScript
- Tailwind/Bootstrap
- Semantic Kernel
- RAG
- Qdrant

Data & Testing
- SQL Server
- PostgreSQL
- xUnit
- Testcontainers
- Swagger/OpenAPI
- Serilog

Education

2020 – 2024 **Bachelor of Languages and translation, English department**

Aswan, Egypt *Al-Azhar University*

Languages

Arabic
Native

English
Excellent